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BrainLift: The GT School Counterfactual — Separating Program Effect from Selection Effect

Owners

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Purpose

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The purpose of this BrainLift is to design a **credible counterfactual** — a pre-registered evaluation a skeptical methodologist would accept — that separates the **program effect** of GT School (the "2-hour" / Timeback acceleration model) from the **selection effect** of admitting students who were already destined to succeed. The anchor literature is SMPY (Study of Mathematically Precocious Youth). Our motivating hypothesis is that GT School's admissions filter — a single CogAT cutoff *plus* a private-school tuition — is **too restrictive**: it selects on demonstrated ability *and* family advantage at once, precisely the profile that makes results dismissible as "smart kids were already smart," and it excludes the population that could actually demonstrate a *real* program effect: **capable-but-underserved** students who are genuinely able (on broader measures than one narrow test — spatial reasoning, conscientiousness, demonstrated achievement) but were missed by the test or priced out by the tuition. *(Our initial hunch centered on "motivation," but the evidence refined it: broad "motivation" — grit, growth mindset — is a weak predictor, whereas*

*prior ability, ability breadth, and conscientiousness are the real levers. The thesis is therefore about broadening access on **capability, not lowering the bar.**)* A design — specifically a **capability-gated admission lottery** — that admits such students and still moves them toward elite readiness would be the single cleanest proof that the program, not the selection, is doing the work.

In Scope

- Causal-identification strategies for education programs: regression discontinuity (RD), matched-comparison and propensity-score designs, and randomized/lottery designs
- SMPY **as a study design** — its selection mechanism, cohort structure, comparison-group sub-analyses, and its own selection-bias limitations (not merely its outcome findings)
- The statistical threats that make single-group top-percentile results misleading: selection bias, regression to the mean, and the local (LATE) nature of RD estimates
- What actually predicts achievement growth in high-ability students — prior ability (which keeps predicting with no ceiling), conscientiousness/self-discipline, and ability breadth (incl. spatial reasoning) — and why broad **motivation** constructs (grit, growth mindset) turn out to be weak predictors
- GT School / Alpha School's actual model, its self-reported outcome claims, its admissions/selection criteria, and the public selection-effect critique
- What must be **pre-registered** (population, cutoff, comparison group, primary outcomes, analysis) for the design to survive a hostile methodologist

Out of Scope

- A full literature survey of acceleration typologies and outcomes (covered in the prior draft; referenced here only where it feeds the design)
- General gifted-education equity policy beyond what informs the selection/counterfactual argument
- Engineering/product details of the Timeback platform itself
- Advocacy for or against GT School; the goal is a design that could *falsify* the program's claims, not defend them

DOK 4: Spiky Points of View (SPOVs)

SPOV 1: A capability-gated admission *lottery* is the only way GT School can produce credible proof that its program, not its selection, drives its results.

- **Elaboration:** GT's core problem is that its front door *is* the confound: by admitting only students who clear a high CogAT cutoff **and** pay private-school tuition (Insight 1). The obvious "fix" of *lowering* the CogAT cutoff would genuinely damage GT (Insight 5). The resolution is to broaden **access, not the bar**: admit a small, *randomized* cohort of **capable-but-underserved** students each year, kids identified as genuinely able through broader measures than a single narrow test (spatial reasoning, conscientiousness, demonstrated achievement - Insight 2), who were missed only because that test is too narrow or because they couldn't pay. A lottery among these applicants is the *only* design that severs program effect from both ability and motivation (Insight 6, 3.9), and it is exactly the population where a real program effect is largest and most detectable (Insight 3, 3.7). The usual "a small school's lottery is underpowered" objection collapses on GT's own terms: if its effects are anywhere near as large as it advertises, a sparse cohort, pooled across years on the incoming voucher ramp (5.5), is more than enough to detect them (Insight 6).

SPOV 2: The same lottery that fixes GT School's evidence problem also bridges the racial and class "excellence gap" its current screen widens because the equity failure and the measurement failure are one and the same.

- **Elaboration:** The same culturally and income-loaded CogAT-plus-tuition filter that confounds GT's data is exactly what excludes capable Black, Hispanic, low-income, and ELL students, so the equity failure and the measurement failure are the same failure (Insight 4, Cat 6), meaning a capability-gated lottery bridges the racial and class excellence gap in the very act of producing the counterfactual, handing an opportunity to able students on the basis of capability rather than privilege.

SPOV 3: If those capable-but-underserved students keep pace, GT School should *expand* the lottery, turning a one-off proof into its fastest route from an advantaged-only boutique to a genuine educational powerhouse.

- **Elaboration:** Expansion is nearly free because GT's academics are self-paced software: the added students barely touch the fast track, and the only real cost sits in shared workshops and guide time which are manageable design choices, not reasons to keep the gate shut (Insight 5, 5.6). So if those students keep pace (3.7), each expansion of the lottery strengthens the evidence and widens GT's reach at once. This is the path from a boutique for the already-advantaged into a genuine educational powerhouse.
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Experts (Sources)

*Curated for **productive disagreement** — the tension between these people is the raw material for our DOK 3 insights. Grouped by the role each plays in the argument. This section also serves as our outreach shortlist.*

- **Expert 1: David Lubinski & Camilla P. Benbow — Vanderbilt University (SMPY directors)**
 - **Who:** Directors of the Study of Mathematically Precocious Youth, the 50-year longitudinal study of intellectually precocious youth founded by Julian Stanley (1971).
 - **Focus:** Ability measured young (SAT at age 12–13) has strong, *graded* predictive validity for elite STEM outcomes — even *within* the top 1%. Their "beyond the threshold" position: ability and drive are *both* independently necessary; ability does not stop mattering above a cutoff.
 - **Why they matter:** They ran the anchor study and know its design seems better than anyone. Their honest answer to "does early motivation add predictive power *over* early ability?" is a direct test of our thesis. They represent the **ability-is-highly-predictive** pole.
 - **Where:** <https://my.vanderbilt.edu/smpy/>
- **Expert 2: Joshua Angrist & Parag Pathak — MIT Blueprint Labs**
 - **Who:** Economists (Angrist is a Nobel laureate for causal-inference methods); co-authors of *The Elite Illusion*. Blueprint Labs partners directly with school systems on admissions-based quasi-experiments.

- **Focus:** Regression discontinuity around selective-school admission cutoffs. Their exam-school work finds that the dramatic peer/environment upgrade at the cutoff has **little causal effect on achievement** — the apparent advantage of elite schools is largely selection, not value-added.
- **Why they matter:** They are the skeptical methodologist our design must satisfy. If our identification strategy survives their critique, it survives anyone. They represent the **selection-dominates-at-the-margin** pole.
- **Where:** <https://blueprintlabs.mit.edu/>
- **Expert 3: Scott Imberman — Michigan State University**
 - **Who:** Co-author of Bui, Craig & Imberman (2014), the closest analog to our exact setting: an RD (plus magnet lottery) evaluation of gifted-and-talented programs.
 - **Focus:** Found that marginal gifted-program admits, despite higher-achieving peers and a more advanced curriculum, show **near-zero achievement gains** in the short run.
 - **Why they matter:** His null result is the benchmark GT School must beat. We would ask him what evidence would convince him a program like GT School produces a real effect.
 - **Where:** <https://www.aeaweb.org/articles?id=10.1257/pol.6.3.30>
- **Expert 4: David Card & Laura Giuliano — UC Berkeley / University of Miami**
 - **Who:** Economists (Card is a Nobel laureate); authors of the universal-screening study for gifted identification.
 - **Focus:** When a district screened *all* students instead of relying on referrals, gifted identification of disadvantaged and minority students rose sharply — the traditional filter was systematically *missing* qualified students.
 - **Why they matter:** This is empirical support for our core intuition that conventional gifted selection is **too restrictive** and misses students who belong. The "hidden gifted" they surface are cousins of our "motivated but behind" cohort.
 - **Where:** <https://www.pnas.org/doi/10.1073/pnas.1605043113>
- **Expert 5: Angela Duckworth — University of Pennsylvania**

- **Who:** Psychologist; originator of "grit" (perseverance and passion for long-term goals) and the Grit Scale.
- **Focus:** Grit predicts achievement and retention (West Point, National Spelling Bee, GPA) over and above IQ/SAT — the case that a non-cognitive trait can be a legitimate selection signal.
- **Why they matter:** The **pro-motivation-as-a-selection-criterion** pole. If motivation is admissible as an admissions signal, her framework is our starting point.
- **Where:** <https://angeladuckworth.com/>
- **Expert 6: Marcus Credé — Iowa State University**
 - **Who:** Psychologist; lead author of the "Much Ado About Grit" meta-analysis.
 - **Focus:** Grit correlates only weakly with performance ($\rho \approx .18$) and is nearly indistinguishable from plain conscientiousness ($\rho \approx .84$); grit interventions have weak effects.
 - **Why they matter:** The **skeptic** we need so the design isn't built on sand. If we select on "motivation," he forces the question: *what are we actually measuring, and how do we keep it reliable and un-gameable?*
 - **Where:** <https://pubmed.ncbi.nlm.nih.gov/27845531/>
- **Expert 7: David Yeager — University of Texas at Austin**
 - **Who:** Psychologist; led the National Study of Learning Mindsets (published in *Nature*, 2019).
 - **Focus:** A short growth-mindset intervention improved grades **specifically for lower-achieving students**, with little effect on high achievers — a rigorously identified effect concentrated in the "behind" population.
 - **Why they matter:** The strongest external evidence *for* our hypothesis that motivation-relevant interventions do the most work precisely for the students GT School screens out. We would ask him whether a motivation-selected, initially-behind cohort is the ideal population to reveal a real treatment effect.
 - **Where:** <https://www.nature.com/articles/s41586-019-1466-y>
- **Expert 8: Pamela Hobart — GT School "gifted evangelist" / Above Grade Level**

- **Who:** Columbia Teachers College–trained gifted-education advocate; the philosophical voice of GT School.
 - **Focus:** Critiques public schooling's substitution of social-emotional programming for academic challenge; articulates the operational case for the GT School model.
 - **Why they matter:** The insider who can explain *why* GT School selects the way it does — and what happens to a highly motivated applicant who is two grades behind. Her rationale is the thing our thesis contests.
 - **Where:** <https://abovegradelevel.substack.com/>
 - **Expert 9: Dylan Kane — education writer (*Five Twelve Thirteen*)**
 - **Who:** Math teacher and education blogger who wrote a widely-read critical analysis of Alpha School's numbers.
 - **Focus:** Argues Alpha should compare students to **their own pre-enrollment trajectory, not the national median**, since \$40k-tuition families are already learning faster than median; credits real fast learning but attributes it to **motivation, not the technology**.
 - **Why they matter:** He has already articulated the exact selection-effect critique our design must neutralize — and independently landed on our "it's motivation" intuition.
 - **Where:** <https://fivetwelvethirteen.substack.com/p/alpha-schools-secret-sauce>
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DOK 3: Insights

1. **Insight 1: GT School's selection process directly hinders the measurement of program effect** GT School only lets in kids who already score high on the CogAT (5.3) and pay a private school level of tuition (5.3, 6.3). Critics claim that this barrier to entry for the GT program leads to ambiguous results for program effect (5.4), as admitted children are selected based on the results used to determine program effect, biasing the statistic in favor of GT. For example, if a gifted kid who passed the CogAT and whose family could afford the tuition chose not to attend GT School, there is very little evidence to suggest that those same financial resources poured

into alternative educational tools for the student couldn't produce the same intellectual boost that GT promised.

2. **Insight 2: CogAT is too narrow of a cognitive assessment to adequately identify the large variety of gifted talent that exists.** CogAT is great at identifying exemplary mathematical and scientific reasoning in children, but struggles with encapsulating other forms of reasoning like spatial reasoning (1.6). Leaving this group of gifted talent out of the program further isolates GT's statistics as selection effect rather than program effect. Additionally, CogAT doesn't account of conscientiousness/self-discipline at all, which is a major factor in a child's categorization as gifted even without top percentile STEM or spatial reasoning (4.5).
3. **Insight 3: The children who could demonstrate GT School's program effect are turned away by the selection process.** As Alpha School has already demonstrated quite well, the TimeBack platform and adaptive learning software see their largest benefits with students who are capable but underserved, not necessarily with the already-thriving (3.7). Since the CogAT cutoff and tuition barrier only admits students with high test scores & adequate resources (5.3), the group that could truly prove their acceleration is a result of GT School and its program are screened out. In scientific terms, this group of capable, but underserved students are being moved from the control situation of traditional schooling and average learning opportunities to GT School, isolating the program effect to just GT School. In contrast, the already-thriving group do not start at the control, as their educational journey was likely to be allocated towards gifted programs regardless of if they got into GT School, making it difficult to isolate the program effect from the student's abilities.
4. **Insight 4: GT School's racial and class bias is not a separate ethical issue — it is the engine of its selection-effect problem.** The CogAT and the tuition don't exclude capable kids at random; they exclude them along racial and class lines. Aptitude scores track family income from as early as age 2 (6.3), the tests are not culturally neutral: English-language learners and culturally diverse students score meaningfully lower on the same instruments (6.2) and even at identical achievement, Black and Hispanic students are identified as gifted far less often (6.1). Add a private-school tuition on top (6.3) and GT's admitted cohort becomes

demographically homogeneous and overwhelmingly advantaged. That homogeneity is exactly what makes program effect impossible to separate from selection effect (5.4): when nearly everyone admitted comes from a high-resource background, there is no contrast left to test whether GT, rather than the background, produced the outcome. The same two filters that make GT inequitable are what pack its applicant pool with already-advantaged students, so the equity failure and the measurement failure are the same failure and the disparity fixes the field already recommends (universal screening, local norms, multiple measures - 6.4) are the same moves that would broaden the pool enough to make a real program-vs-selection comparison possible.

5. Insight 5: Lowering the CogAT cutoff would genuinely dilute GT's edge, a capability-based lottery would not. GT and its sister school Alpha run the same 2-hour/Timeback model (5.1); the only difference is audience: Alpha serves a broad ability range, GT is gifted-only (5.1, 5.3). That exclusivity is GT's real edge because it can aim its entire shared environment (workshops, guides, norms, pace) at the very best and most promising students (5.6), which Alpha and broad-range gifted programs cannot. So if GT broadened access by *lowering the CogAT cutoff* (the direction local-norm / lower-threshold remedies point - 6.4), it would genuinely dilute that edge, admitting lower-ability students and dragging the shared environment toward the middle. But a lottery/voucher among *capable-but-underserved* students is a different move: it selects on capability, not privilege, giving an opportunity to kids who are just as able but were missed by a narrow test or priced out by tuition (Insights 2–4). It broadens *access*, not the *bar*. The dilution GT rightly fears comes from lowering the threshold, not from admitting capable students through a fairer door.

6. Insight 6: An admission lottery is the one tool that could prove GT works and because GT claims such huge effects, it wouldn't take many students to do it. A lottery among applicants is the only design that separates program effect from *both* ability and motivation, because winners and losers are identical on average (3.9). Programs that run lotteries show real gains (KIPP ~0.35 SD/yr), largest for kids who started behind (3.9, 3.7). The usual objection: a single school's lottery is too small to detect anything (3.9) collapses here. GT advertises enormous effects (2.6x growth), and large effects need only small samples to detect (3.9). So even a sparse cohort, pooled across a few years (which fits the voucher ramp - 5.5), is

enough to test GT's own claim. The lottery isn't a threat to GT; it's the cheapest proof it could offer.

DOK 2: Knowledge Tree

*Broad categories → sources → **DOK 1 facts** (what two analysts would extract identically) and a **DOK 2 summary** (compression in our own words, each sentence traceable to a fact). Verification flags for load-bearing citations are collected in the appendix.*

Category 1: Why GT School's CogAT Screen May Be Too Strict — The False-Negative Problem in Cognitive Selection

This category assembles the education-specific, statistical evidence that a single cognitive-ability screen (CogAT at ~90th percentile) systematically misses students who could reach elite achievement (1.1–1.4), the steelman counterargument for why cognitive screening is nonetheless defensible (1.5), and a further, purely ability-based false negative — the under-selection of spatial talent (1.6). The formal causal-inference machinery that justifies the design itself is in Category 3.6; the distributional consequences of the screen are in Category 6.

- **Subcategory 1.1: Highly correlated tests identify different students — one screen misses most of the "best" kids**
 - **Source:** Lohman (2005); Lohman & Gambrell (2012)
 - **DOK 1 — Facts:**
 - Lohman, D. F. (2005), "An Aptitude Perspective on Talent: Implications for Identification of Academically Gifted Minority Students," *Journal for the Education of the Gifted*, 28(3/4), 333–360 — using the joint ITBS/CogAT national standardization (~14,000 students/grade, grades 3–6): of students in the **top 3% on ITBS Reading**, the **CogAT Composite identifies only 32%**, the CogAT Verbal battery 35%, and the **CogAT Nonverbal battery only 18%** (even the ITBS Composite catches just 54%).
 - Lohman (2005) states: "correlations generally imply far less agreement between scores than most people think, especially for extreme scores."

(CogAT Composite $\leftrightarrow$ ITBS Reading $r = .79$; CogAT Nonverbal $\leftrightarrow$ Reading $r = .62$.)

- Lohman, D. F., & Gambrell, J. L. (2012), "Using Nonverbal Tests to Help Identify Academically Talented Children," *Journal of Psychoeducational Assessment*, 30(1), 25–44 — "large discrepancies between even highly correlated abilities become increasingly common at the extremes"; CogAT Nonverbal SEM ≈ 3.7 (scale SD = 16), so a student's true-score band straddles any single cutoff.
- **DOK 2 — Summary:** A single cognitive screen does not identify "the" able students; it identifies the subset that one test happens to rank highly, and at the extreme tail even tests correlated .6–.8 disagree sharply. On these numbers a CogAT-composite cutoff at the 90th percentile would pass over the majority of children who are genuinely top performers on an achievement criterion, and measurement error around the cutoff makes any single pass/fail line partly arbitrary.
- **Link to source:** <https://files.eric.ed.gov/fulltext/EJ746059.pdf> | <https://library.scottbarrykaufman.com/uploads/2013/04/Lohman-Gambrell-2012.pdf>
- **Subcategory 1.2: A one-time childhood score is a moving target — early rank does not predict later rank**
 - **Source:** Lohman & Korb (2006) / Martin (1985); Ramsden et al. (2011); Kelley's regression-to-the-mean (Maassen 2000)
 - **DOK 1 — Facts:**
 - Lohman, D. F., & Korb, K. A. (2006), "Gifted Today but Not Tomorrow? Longitudinal Changes in Ability and Achievement During Elementary School," *Journal for the Education of the Gifted*, 29(4), 451–484 — of students in the **top 3% in mathematics at grade 3, ~70% were no longer in the top 3% by grade 8** (grade 3 $\rightarrow$ 8 correlation $r = .73$); only ~40% of top-3% grade-3 composite scorers were still top-3% one year later at grade 4.
 - Ramsden, S., et al. (2011), "Verbal and non-verbal intelligence changes in the teenage brain," *Nature*, 479, 113–116 — individual IQ shifted by **–20 to +23 points (verbal)** and –18 to +21 (full-scale) over ~3.5 years; ~33% of

teens changed full-scale IQ beyond the 90% confidence interval; 21% shifted verbal IQ ≥ 15 points (≥ 1 SD). The changes tracked structural brain changes (real, not measurement noise).

- Regression to the mean: because any test has imperfect reliability, a group selected for extreme scores drifts toward the mean on retest by an amount proportional to $(1 - \text{reliability})$ — the statistical mechanism behind the fall-out above (Kelley's formula; verified anchor: Maassen, G. H. (2000), *Psychometrika*, 65(2), 187–197).
- **DOK 2 — Summary:** The rank a child holds on a one-time test in early childhood is a weak guide to the rank they will hold years later — both because ability itself changes through adolescence and because extreme early scores partly reflect measurement luck that regresses. An admission screen applied once, at entry, locks in a snapshot that a large fraction of eventual high achievers would fail at that moment even if they would clear it later.
- **Link to source:** <https://files.eric.ed.gov/fulltext/EJ746292.pdf> | <https://www.nature.com/articles/nature10514> | <https://doi.org/10.1007/BF02294373>
- **Subcategory 1.3: The gatekeeping stage itself manufactures false negatives**
 - **Source:** McBee, Peters & Miller (2016); McBee, Peters & Waterman (2014); Peters et al. (2019)
 - **DOK 1 — Facts:**
 - McBee, M. T., Peters, S. J., & Miller, E. M. (2016), "The Impact of the Nomination Stage on Gifted Program Identification," *Gifted Child Quarterly*, 60(4), 258–278 — simulation shows that requiring a screening/nomination gate before testing means "a large proportion of gifted students being missed"; under commonly implemented conditions the **false-negative rate can easily exceed 60%**. A screening stage can only lose students, never recover them.
 - McBee, M. T., Peters, S. J., & Waterman, C. (2014), "Combining Scores in Multiple-Criteria Assessment Systems: The Impact of Combination Rule," *Gifted Child Quarterly*, 58(1), 69–89 — the combination rule drives who qualifies: an "and" (conjunctive) rule minimizes the identified pool and

maximizes false negatives; the compensatory "mean" rule is the most accurate.

- Peters, S. J., et al. (2019), "Effect of Local Norms on Racial and Ethnic Representation in Gifted Education," *AERA Open*, 5(2) — switching from a single national cutoff to local norms changes which students qualify.
- **DOK 2 — Summary:** Even holding the test fixed, the *structure* of a selection process — a hard gate, a conjunctive "must clear the bar on this one test" rule, a single national cutoff — is itself a large source of false negatives, independent of any child's true ability. GT School's single-CogAT-cutoff gate is precisely the configuration this literature identifies as maximizing missed students.
- **Link to source:**
<https://journals.sagepub.com/doi/abs/10.1177/0016986216656256> |
<https://journals.sagepub.com/doi/10.1177/0016986213513794> |
<https://journals.sagepub.com/doi/10.1177/2332858419848446>
- **Subcategory 1.4: Documented talent loss — the pool feeding the screen is already leaking, hardest for the disadvantaged**
 - **Source:** Wyner, Bridgeland & Dilulio (2007), *Achievement Trap*; Plucker et al. (excellence gap)
 - **DOK 1 — Facts:**
 - Wyner, J. S., Bridgeland, J. M., & Dilulio, J. J. (2007), *Achievement Trap: How America Is Failing Millions of High-Achieving Students from Lower-Income Families*, Jack Kent Cooke Foundation & Civic Enterprises — "more than 3.4 million K-12 students achieving in the top quartile academically come from families earning less than the median income."
 - *Achievement Trap*: of lower-income students in the top reading quartile as first graders, only 56% remained in it by fifth grade (vs. 69% of higher-income) — i.e., ~44% fell out vs. ~31%.
 - *Achievement Trap*: among students *not* in the top quartile in first grade, only ~7% of lower-income students rose into it by fifth grade (math and reading), vs. 16–17% of higher-income students — lower-income high-potential students rarely rise in; the report attributes the loss to low-expectation school environments, not lack of ability.

- Excellence-gap data (Plucker et al.): on NAEP Grade 4 math, roughly 3% of lunch-assistance students reached the "Advanced" tier vs. ~14% of others, a gap that has persisted across administrations.
- **DOK 2 — Summary:** Before GT School ever applies its screen, the population of "already-demonstrated" high performers has been thinned by years of unequal environments — able students, disproportionately lower-income, fall out of or never enter the measured top tier, and the report ties this to opportunity, not ability. A screen that selects on demonstrated standing therefore inherits that prior loss, systematically excluding exactly the students whose ceiling was never fairly tested.
- **Link to source:** <https://www.jkcf.org/research/achievement-trap-how-america-is-failing-millions-of-high-achieving-students-from-lower-income-families/> | <https://fordhaminstitute.org/national/commentary/achievement-trap-how-america-failing-millions-high-achieving-students-lower>
- **Subcategory 1.5: Counterargument — the case FOR cognitive screening (steelman)**
 - **Source:** Deary et al. (2007); Card & Giuliano (2016); Grissom & Redding (2016); Lohman (2005); Makel et al. (2016); Hobart (in Segan / Reason 2026)
 - **DOK 1 — Facts:**
 - Deary, I. J., Strand, S., Smith, P., & Fernandes, C. (2007), "Intelligence and educational achievement," *Intelligence*, 35(1), 13–21 — in a prospective study of >70,000 children, a cognitive test (CogAT) at age 11 correlated $r \approx .69$ (**observed; .81 latent**) with GCSE educational achievement at 16; g explained ~59% of variance in mathematics. Cognitive ability is among the strongest single early predictors of later achievement.
 - Card & Giuliano (2016, PNAS): replacing referral with **universal** cognitive screening raised gifted identification of disadvantaged students **+174%**, Hispanic **+118%**, Black **+74%** — an objective test, applied to everyone, surfaced far *more* underrepresented talent than human referral. (Same study appears in 3.3 and Category 6.)
 - Grissom, J. A., & Redding, C. (2016), "Discretion and Disproportionality," *AERA Open*, 2(1) — at identical achievement, Black students were assigned

to gifted programs about *half* as often as White peers; a test-based rule removes some of that discretionary human bias.

- Lohman (2005) is a *proponent* of aptitude testing done well — using local norms and multiple measures — not an opponent of testing per se.
- Makel, M. C., Kell, H. J., Lubinski, D., Putallaz, M., & Benbow, C. P. (2016), "When Lightning Strikes Twice: Profoundly Gifted, Profoundly Accomplished," *Psychological Science*, 27(7), 1004–1018 — above-level ability testing at age 12 predicted exceptional attainment (doctorates, tenure, patents) by ~age 40, validating early aptitude identification.
- GT School's documented position, via Pamela Hobart (in Segan, *Reason*, 2026): "the way you actually find very high-ability learners falling through the cracks is you just standardized-test everyone"; she argues moving to "multiple measures" risks omitting high-IQ students who don't display leadership or aren't nominated. (*No documented statement from Joe Liemandt or MacKenzie Price crediting CogAT specifically with a positive impact was located — see appendix.*)
- **DOK 2 — Summary:** The honest counterweight to 1.1–1.4 is that cognitive-ability testing is among the best-validated single predictors of achievement, and that a standardized test applied *universally* is more equitable than the subjective referral it replaces — the Card & Giuliano result shows the test *widened* access rather than narrowing it. Notably, the defensible pro-screening positions on record (Lohman; Hobart) do not defend a single fixed cutoff but rather universal testing combined with multiple measures — which locates the real disagreement at *single-cutoff gate vs. universal-screen-plus-multiple-measures*, not test vs. no test.
- **Link to source:** <https://doi.org/10.1016/j.intell.2006.02.001> | <https://www.pnas.org/doi/10.1073/pnas.1605043113> | <https://doi.org/10.1177/2332858415622175> | <https://doi.org/10.1177/0956797616644735>
- **Subcategory 1.6: The Screen Misses an Entire Ability Dimension — Spatial Talent**
 - **Source:** Wai, Lubinski & Benbow (2009); Kell & Lubinski (2014)
 - **DOK 1 — Facts:**

- Wai, J., Lubinski, D., & Benbow, C. P. (2009), "Spatial Ability for STEM Domains: Aligning Over 50 Years of Cumulative Psychological Knowledge Solidifies Its Importance," *Journal of Educational Psychology*, 101(4), 817–835 — on Project Talent (**N > 400,000**, 11-year longitudinal tracking): spatial ability is a robust predictor of STEM achievement and attainment *over and above* math and verbal ability.
- Talent identification restricted to math/verbal measures **misses roughly 50% of the top 1% in spatial ability**; approximately **70% of the spatial top 1% do not fall in the math or verbal top 1%** (Lubinski/Benbow analyses; corroborated in the SMPY top-0.01% sample).
- Ability *level* predicts the *magnitude* of achievement; ability *pattern* (a spatial>verbal tilt plus math) predicts its *domain* (STEM).
- **DOK 2 — Summary:** Even granting the case for cognitive screening (1.5), a verbal/quant-weighted composite systematically overlooks an entire, well-validated ability dimension: spatial reasoning. Because roughly half of the spatially most able students are invisible to math/verbal talent searches, a CogAT-style cutoff misses a large pool of exactly the profile most predictive of STEM attainment — a false negative that is about *ability breadth*, not motivation, and one a multi-measure screen (including spatial) would recover.
- **Link to source:** <https://my.vanderbilt.edu/smpy/> | <https://gwern.net/doc/iq/high/smpy/2014-kell.pdf>

Category 2: SMPY as a Design (the anchor literature)

- **Subcategory 2.1: The Selection Mechanism & Cohort Structure**
 - **Source:** Lubinski & Benbow (2006)
 - **DOK 1 — Facts:**
 - SMPY founded by Julian C. Stanley, Sept. 1, 1971, Johns Hopkins; selection via *above-level* testing — administering the SAT to 12–13-year-olds to discriminate "the able from the exceptionally able."
 - Five cohorts, >5,000 participants, identified 1972–1997; planned 50-year longitudinal design (follow-ups at ages 18, 23, 33, 50, 65). Now directed by David Lubinski & Camilla Benbow at Vanderbilt.

- Cohort cutoffs (top-X-in-ability by age 13): Cohort 1 (1972–74) $\approx$ top 1% (SAT-M $\geq$ 390 or SAT-V $\geq$ 370); Cohort 2 $\approx$ top 0.5% (SAT-M $\geq$ 500 or SAT-V $\geq$ 430); Cohort 3 $\approx$ top 0.01% (SAT-M $\geq$ 700 or SAT-V $\geq$ 630); Cohort 4 $\approx$ top 3% screen; Cohort 5 = graduate students in top-15 STEM programs (not SAT-selected).
 - **DOK 2 — Summary:** SMPY's power comes from a sharp, quantified selection rule (a single above-level SAT at 12–13) applied to talent-search volunteers, producing cohorts stratified by ability level. The design is a prospective longitudinal tracking of a *selected* pool — the selection is transparent, but **participants first self-selected into talent searches**, so the sample is not a random draw of high-ability youth.
 - **Link to source:** <https://journals.sagepub.com/doi/10.1111/j.1745-6916.2006.00019.x> | <https://my.vanderbilt.edu/smpy/>
- **Subcategory 2.2: Comparison-Group Sub-Analyses (SMPY's own counterfactual moves)**
 - **Source:** Swiatek & Benbow (1991); Lubinski & Benbow (2006) within-cohort analyses
 - **DOK 1 — Facts:**
 - Swiatek, M. A., & Benbow, C. P. (1991), "A ten-year longitudinal follow-up of ability-matched accelerated and unaccelerated gifted students," *Journal of Educational Psychology*, 83, 528–538 — an **ability-matched** accelerated-vs-non-accelerated comparison (quasi-experimental, not randomized).
 - Lubinski & Benbow (2006) report within-top-1% dose-response: splitting participants into top vs. bottom SAT-M quartiles yields divergent outcomes — ability differences *inside* the top 1% still predict divergence.
 - Reported gradient: ~30% doctorate attainment for those scoring $\geq$ 500 SAT-M at 13 vs. ~50% for those $\geq$ 700.
 - **DOK 2 — Summary:** SMPY never randomized anyone to "gifted" status or to acceleration; its comparisons rely on *matching* accelerated to non-accelerated students of equal measured ability. Matching controls measured ability but leaves motivation and family factors uncontrolled — the same limitation any matched GT School design will face.

- **Link to source:** <https://journals.sagepub.com/doi/10.1111/j.1745-6916.2006.00019.x>
- **Subcategory 2.3: Predictive-Validity Findings (ability measured young)**
 - **Source:** Kell, Lubinski & Benbow (2013); Bernstein, Lubinski & Benbow (2019)
 - **DOK 1 — Facts:**
 - Kell, H. J., Lubinski, D., & Benbow, C. P. (2013), "Who Rises to the Top? Early Indicators," *Psychological Science*, 24(5), 648–659 — N=320 identified before age 13 in the top 0.01%, assessed by age 38: 44% held doctorates (vs. ~2% U.S. base rate), ~15% held ≥1 patent (vs. ~1%), >11% held university tenure.
 - Bernstein, B. O., Lubinski, D., & Benbow, C. P. (2019), "Psychological Constellations Assessed at Age 13 Predict Distinct Forms of Eminence 35 Years Later," *Psychological Science*, 30(3), 444–454 — ability/preference profiles at 13 predicted distinct forms of eminence at ~50.
 - **DOK 2 — Summary:** SMPY is the strongest evidence that a single ability measure taken young has real, long-range predictive validity, and that differences persist even within the extreme tail. This is the empirical case *against* our thesis — it indicates that admitting on ability really does forecast who succeeds — and any counterfactual design must confront it head-on rather than wish it away.
 - **Link to source:** <https://pubmed.ncbi.nlm.nih.gov/23531483/> | <https://pmc.ncbi.nlm.nih.gov/articles/PMC6419263/>
- **Subcategory 2.4: SMPY's Own Selection-Bias Limitations (design honesty)**
 - **Source:** Lubinski & Benbow (2006), authors' own caveats
 - **DOK 1 — Facts:**
 - On an AP-participation finding, the authors state the effect could operate "through self-selection or something intrinsic to the AP program itself" — i.e., they explicitly cannot separate program from selection.
 - Identification rests on a single SAT administration at 12–13; no random assignment to gifted status or to acceleration.
 - Cohorts skew by geography (Maryland, mid-Atlantic, Midwest) and era (pre-recentered SAT), limiting representativeness.

- **DOK 2 — Summary:** The authors of the anchor study concede its central limitation in print: an observational design that selects on ability and lets participants self-select into programs cannot cleanly attribute outcomes to the program. That admission is the seam our credible-counterfactual argument can pry open — the design we propose should do the one thing SMPY could not. (*The studies that do attribute growth to a program net of selection — Card & Giuliano; Booij — are in 3.7; they are the causal move SMPY structurally cannot make.*)
- **Link to source:** <https://journals.sagepub.com/doi/10.1111/j.1745-6916.2006.00019.x>

Category 3: Quasi-Experimental Designs That Separate Program from Selection

- **Subcategory 3.1: Regression Discontinuity at Admission Cutoffs (the exam-school evidence)**
 - **Source:** Abdulkadiroğlu, Angrist & Pathak (2014); Dobbie & Fryer (2014); Bui, Craig & Imberman (2014)
 - **DOK 1 — Facts:**
 - Abdulkadiroğlu, A., Angrist, J., & Pathak, P. (2014), "The Elite Illusion: Achievement Effects at Boston and New York Exam Schools," *Econometrica*, 82(1), 137–196 — fuzzy RD at exam-school cutoffs; large peer-quality jumps at the cutoff have "little causal effect on test scores or college quality."
 - Dobbie, W., & Fryer, R. (2014), "The Impact of Attending a School with High-Achieving Peers: Evidence from the New York City Exam Schools," *AEJ: Applied Economics*, 6(3), 58–75 — just-eligible NYC applicants get peers 0.17–0.36 SD higher and 6.4–9.5 pp less likely to be Black/Hispanic, yet see little impact on achievement or college outcomes.
 - Bui, S., Craig, S., & Imberman, S. (2014), "Is Gifted Education a Bright Idea? Assessing the Impact of Gifted and Talented Programs on Students," *AEJ: Economic Policy*, 6(3), 30–62 — RD at gifted-eligibility cutoffs plus a magnet lottery; marginal gifted admits' achievement "neither improves nor worsens" short-run; only science improves in the lottery arm.

- **DOK 2 — Summary:** The best-identified studies of selective/gifted schooling repeatedly find that the enormous apparent advantage of elite programs is mostly *who gets in*, not *what happens inside* — at the admission margin, the causal effect on achievement is close to zero. This is both the template for a GT School RD and its cautionary tale: the null at the margin is the result GT School would have to beat.
- **Link to source:** <https://onlinelibrary.wiley.com/doi/abs/10.3982/ECTA10266> | <https://www.aeaweb.org/articles?id=10.1257/app.6.3.58> | <https://www.aeaweb.org/articles?id=10.1257/pol.6.3.30>
- **Subcategory 3.2: RD Identifies a *Local* Effect (the LATE limitation)**
 - **Source:** Imbens & Lemieux (2008); Lee & Lemieux (2010)
 - **DOK 1 — Facts:**
 - Imbens, G., & Lemieux, T. (2008), "Regression discontinuity designs: A guide to practice," *Journal of Econometrics*, 142(2), 615–635.
 - Lee, D., & Lemieux, T. (2010), "Regression Discontinuity Designs in Economics," *Journal of Economic Literature*, 48(2), 281–355.
 - RD estimates a Local Average Treatment Effect — the effect only for students *at the cutoff* — and is silent about *inframarginal* (well-above-cutoff) students.
 - **DOK 2 — Summary:** An RD around GT School's CogAT cutoff would answer "does GT School help the *marginal* admit?" — not "does it help its typical top student?" That distinction is decisive: a null at the margin (as the exam-school studies find) does not disprove an effect for the inframarginal, and an effect at the margin does not generalize upward. Whether RD is even the right primary design here hinges on which population we most care about.
 - **Link to source:** <https://doi.org/10.1016/j.jeconom.2007.05.001> | <https://www.aeaweb.org/articles?id=10.1257/jel.48.2.281>
- **Subcategory 3.3: Universal Screening — Selection Is Too Restrictive**
 - **Source:** Card & Giuliano (2016)
 - **DOK 1 — Facts:**
 - Card, D., & Giuliano, L. (2016), "Universal screening increases the representation of low-income and minority students in gifted education,"

PNAS, 113(48), 13678–13683.

- A large Florida district switched from referral-based to *universal* second-grade screening (NNAT); odds of gifted identification rose ~45% overall, with increases of ~+118% (Hispanic), +74% (Black), and +174% (disadvantaged, under a lower IQ ≥ 116 threshold) — with no change in eligibility standards.
- Pre-program, 13 schools had zero gifted third-graders; post-program all 140 had ≥ 1 .
- **DOK 2 — Summary:** When the *filter* changed but the *bar* did not, the system found large numbers of qualified gifted students it had previously missed — disproportionately poor and minority. This is direct evidence that conventional gifted selection is systematically too narrow, and it supports the premise that GT School's ability screen may exclude admissible students who would succeed if let in.
- **Link to source:** <https://www.pnas.org/doi/10.1073/pnas.1605043113>
- **Subcategory 3.4: Matched-Comparison & Propensity-Score Designs (and their ceiling)**
 - **Source:** Rosenbaum & Rubin (1983); Steenbergen-Hu, Makel & Olszewski-Kubilius (2016)
 - **DOK 1 — Facts:**
 - Rosenbaum, P., & Rubin, D. (1983), "The Central Role of the Propensity Score in Observational Studies for Causal Effects," *Biometrika*, 70(1), 41–55 — matching on the propensity score removes bias due to *observed* covariates; requires strong ignorability (no unmeasured confounding).
 - Steenbergen-Hu, S., Makel, M., & Olszewski-Kubilius, P. (2016), "What One Hundred Years of Research Says About the Effects of Ability Grouping and Acceleration...," *Review of Educational Research*, 86(4), 849–899 — acceleration vs. same-age non-accelerated peers $g \approx 0.70$; vs. older non-accelerated peers $g \approx 0.09$ (n.s.).
 - **DOK 2 — Summary:** Matching and propensity-score designs can build an ability-comparable control group, but they only balance what is *measured* — the unobserved drivers we most worry about (motivation, parenting, family

investment) stay uncontrolled, which is exactly the "smart kids were already smart" residual. The acceleration meta-analysis also shows the comparison-group choice (same-age vs. older peers) swings the effect from large to null, underscoring how much the counterfactual definition drives the answer.

- **Link to source:** <https://academic.oup.com/biomet/article-abstract/70/1/41/240879> | <https://doi.org/10.3102/0034654316675417>
- **Subcategory 3.6: The Formal Identification Problem the Design Must Solve (why a counterfactual is required at all)**
 - **Source:** Holland (1986); Rubin (1974); Heckman (1979); Campbell & Stanley (1963)
 - **DOK 1 — Facts:**
 - Holland, P. W. (1986), "Statistics and Causal Inference," *Journal of the American Statistical Association*, 81(396), 945–960 — the "Fundamental Problem of Causal Inference": for any single unit one can observe the outcome under treatment or control, never both; coined "Rubin causal model" and "no causation without manipulation."
 - Rubin, D. B. (1974), "Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies," *Journal of Educational Psychology*, 66(5), 688–701 — a causal effect is the difference between potential outcomes.
 - Heckman, J. J. (1979), "Sample Selection Bias as a Specification Error," *Econometrica*, 47(1), 153–161 — non-random selection into a group is an omitted-variable/specification error that biases estimates.
 - Campbell, D. T., & Stanley, J. C. (1963), *Experimental and Quasi-Experimental Designs for Research* — statistical regression and selection are named threats to internal validity; the one-group pretest–posttest design is explicitly vulnerable.
 - **DOK 2 — Summary:** This is the formal reason "smart kids were already smart" is not a rhetorical jab but a design requirement: one never observes what a GT School student *would have* achieved without GT School, so any single-group result confounds the program with whatever made those students selectable. Every design in this category (RD, lottery, matching) is an

attempt to construct that missing counterfactual credibly; without one, selection on unobservables biases the estimate in an unknown direction.

- **Link to source:** <https://www.jstor.org/stable/2289064> | <https://eric.ed.gov/?id=EJ118470> | <https://www.jstor.org/stable/1912352>

- **Subcategory 3.7: Program Effect Net of Selection — When a Gifted Program *Actually* Raises Achievement (and When It Doesn't)**

- **Source:** Card & Giuliano (2016); Booij, Haan & Plug (2016, 2017); reconciled against the nulls in 3.1

- **DOK 1 — Facts (positive results):**

- Card, D., & Giuliano, L. (2016), "Can Tracking Raise the Test Scores of High-Ability Minority Students?", *American Economic Review*, 106(10), 2783–2816 — rank-based **RD** (n=4,144) on a separate high-achiever classroom entered by *prior-achievement* rank: full sample **+0.29 SD reading / +0.28 math**; Black & Hispanic high-achievers **+0.5–0.7 SD** (Black-only +0.65/+0.69); white/advantaged students **≈ 0**; gains persist to 6th grade ($\sim 0.2\sigma$); no negative spillovers on non-participants.
- Booij, A. S., Haan, F., & Plug, E. (2016), "Enriching Students Pays Off..." (IZA DP 9757) — **fuzzy RD** at a gifted-program aptitude cutoff (N=3,127, one selective Dutch school): GPA **+0.38 SD math / +0.30 language / +0.44 other**; persists into university field choice. Companion paper (2017, IZA DP 10836; ~2,400 students; DiD+RD): **~+0.2 SD** near the cutoff, with gains *growing* further above it.
- Both attribute gains to a genuinely **differentiated curriculum + higher expectations** for underserved high-achievers — not merely to better peers.

- **DOK 1 — Facts (issues with the positive results — stated plainly):**

- Card & Giuliano identifies a **local** effect at the *achievement* cutoff (not GT School's IQ/CogAT margin); the treatment is a **separate tracked classroom**, not an individualized AI-mastery model; the effect is **~zero for white/advantaged students**, so it does not generalize to all high-ability kids; the district is anonymized (widely identified as Broward County, FL).
- Booij (2016/2017) are **IZA working papers**, not confirmed peer-reviewed journal articles at that stage (verify final publication before load-bearing

use); the 2016 result is a **single already-selective school** (external validity); fuzzy-RD depends on bandwidth/first-stage assumptions; the 2017 per-subject SEs were not independently verified.

- Both are **RD/quasi-experimental, not RCTs**; both measure GPA/test scores, **not** "elite readiness"; **neither tests an intensive AI-tutoring/2-hour model** — so they establish that a program effect is *possible* for high-ability students, not that GT School's specific model produces one.
- **DOK 1 — Facts (counter-evidence — the clean nulls, and their issues):**
 - The RD studies in 3.1 — Bui, Craig & Imberman (2014, gifted magnet ≈ 0), Abdulkadiroğlu, Angrist & Pathak (2014, "Elite Illusion" ≈ 0), Dobbie & Fryer (2014, NYC exam schools ≈ 0) — find that **selective admission alone raises nothing** despite large peer-quality jumps.
 - *Issues with the counter-evidence:* each identifies only a **Local Average Treatment Effect at the admission margin** and is silent on inframarginal (well-above-cutoff) students; their treatment is largely "**better peers, same curriculum**," so they test peer-sorting, not differentiated instruction; the "elite illusion" nulls are *precisely estimated* but population-specific (already-high-performing exam-school applicants). They show peer-sorting doesn't work, not that instruction can't.
 - *Weak-identification upper bound:* the acceleration $g \approx 0.70$ (Steenbergen-Hu et al. 2016, 3.4) is **matched/observational only** — selection-inflated, structurally as vulnerable to "already gifted" as SMPY. Treat it as a ceiling, not a program-effect estimate.
- **DOK 2 — Summary:** Two regression-discontinuity studies do the one thing SMPY cannot — attribute achievement growth to a gifted *program* net of selection, at $+0.3$ – 0.5 SD (up to $+0.7$ for underserved minority high-achievers). Read honestly, they prove a program effect is *possible*, not universal: it appears where selection is on **achievement**, the curriculum is genuinely **differentiated**, and the population is **underserved/inframarginal**, and it disappears (the "elite illusion") where a test threshold mainly delivers **better peers to already-advantaged** students. That contrast is the crux for GT School, whose IQ/CogAT gate + peer environment risk the null case, while

its individualized mastery instruction is the ingredient the positive studies credit.

- **Link to source:** <https://www.aeaweb.org/articles?id=10.1257/aer.20150484> | <https://www.iza.org/publications/dp/9757> | <https://ideas.repec.org/p/iza/izadps/dp10836.html>
- **Subcategory 3.8: The Credible Measurement Stack — Isolating Impact from Both Ability *and* Motivation (capstone)**
 - **Source:** Angrist, Imbens & Rubin (1996); McCrary (2008); Oster (2019); VanderWeele & Ding (2017); Chetty, Friedman & Rockoff (2014) w/ Rothstein (2017); Betebenner (2009); Assouline & Lupkowski-Shoplik (2012); WWC Handbook v5.0 (2022); Nosek et al. (2018)
 - **DOK 1 — Facts:**
 - **Design (neutralizing motivation, ranked):** only random **admission lotteries** balance *unobserved* traits including motivation (Angrist, Imbens & Rubin 1996, JASA — IV/LATE); **RD** neutralizes confounders *only locally at the cutoff* and only absent score manipulation (McCrary 2008 density test); **DiD/CITS** require parallel trends; **matching/propensity** balance *observed* covariates only, leaving motivation confounded.
 - **Sensitivity bounds (when randomization is impossible):** report how strong an unmeasured "motivation" confounder must be to overturn the result — Rosenbaum Γ ; Oster (2019) δ ($\delta \geq 1$ = robust); VanderWeele & Ding (2017) E-value.
 - **Outcome metric:** condition on prior achievement (value-added — Chetty, Friedman & Rockoff 2014, contested by Rothstein 2017; or Student Growth Percentiles — Betebenner 2009, which are *descriptive not causal*); reference growth to the student's **own multi-year baseline, not the national median** (MDRC) — but single-school precision is poor (minimum detectable effect ≈ 0.57 – 0.80 SD).
 - **Ceiling fix (gifted-specific):** grade-level tests cap gifted growth; use **above-level or adaptive/high-ceiling instruments** (Assouline & Lupkowski-Shoplik 2012; SMPY rationale) so growth is even measurable. GT School's own metric (NWEA MAP) has documented high-end ceiling limits.

- **Standards the design must clear:** WWC Handbook v5.0 — only a low-attrition **RCT** "meets standards without reservations"; quasi-experiments need **baseline equivalence** (≤ 0.05 SD no adjustment / $0.05-0.25$ SD adjust / >0.25 SD fails); analyze **intent-to-treat**; **pre-register** the primary outcome and analysis (Nosek et al. 2018); run **placebo/falsification tests** for residual selection.

- **DOK 2 — Summary:** Isolating GT School's program effect is not one metric but a *stack*: (1) a lottery (or cutoff RD) to balance ability *and* motivation by design; (2) a high-ceiling/above-level or adaptive test so gifted growth is visible; (3) growth referenced to each student's own baseline, not a national norm; (4) intent-to-treat analysis, pre-registered, within WWC attrition/equivalence limits; and (5) where randomization is impossible, explicit sensitivity bounds stating how much unmeasured motivation it would take to erase the effect. This stack is the skeleton of the pre-registered design this BrainLift exists to produce.

- **Link to source:**
<https://www.tandfonline.com/doi/abs/10.1080/01621459.1996.10476902> |
<https://ideas.repec.org/a/taf/jnlbes/v37y2019i2p187-204.html> |
<https://www.acpjournals.org/doi/abs/10.7326/M16-2607> |
https://ies.ed.gov/ncee/wwc/Docs/referenceresources/Final_WWC-HandbookVer5_0-0-508.pdf |
<https://www.pnas.org/doi/10.1073/pnas.1708274114>

- **Subcategory 3.9: Admission Lotteries — The Cleanest Counterfactual, What They Find, and the Sparseness/Power Trade-off**
 - **Source:** Cullen, Jacob & Levitt (2006); Angrist et al. (KIPP, 2012); Abdulkadiroğlu, Angrist, Narita & Pathak (2017); Weiland et al. (2024)
 - **DOK 1 — Facts:**
 - **Feasible and internally valid:** Cullen, Jacob & Levitt (2006), *Econometrica* — 194 randomized lotteries across 19 oversubscribed Chicago high schools; because lottery winners and losers among applicants are equivalent on average, outcome differences are causally attributable to the school, using pre-existing records.
 - **Effect sizes when the program works:** KIPP lottery (Angrist et al. 2012) ~ 0.35 SD/yr math; Denver (Abdulkadiroğlu et al. 2017) $0.35-0.42$ SD math

— *and effects were largest for students who started behind.* (Full evidence and the program-effect-vs-selection contrast is in 3.7.)

- **The power problem — and the fix:** single small-school lotteries are often underpowered (most KIPP schools lacked enough applicants; null results reflected underpowering, not zero effect). The remedy is **design-based propensity ("DA-type") pooling across lotteries/cohorts**, which grew Denver's usable sample ~5× (from 462 to ~2,300). Accumulating a small lottery across multiple years is a legitimate way to reach adequate power.
- **Large effects need small samples:** statistical power rises with effect size — the larger the true effect, the smaller the N needed to detect it (standard power relationship). A program advertising very large effects can therefore be tested with a comparatively small lottery cohort.
- **The complier ceiling:** even a valid lottery identifies a Local Average Treatment Effect only for *applicants/compliers*, who are systematically more advantaged than the general population (Weiland et al. 2024: complier neighborhood income far exceeded the applicant average). Results speak to the applicant pool, not to non-applicants.
- **DOK 2 — Summary:** An oversubscription admission lottery is the one design that severs program effect from selection on *both* ability and motivation, and it is cheap and fast on existing records. Its known weakness — small single-school samples lack power — is solvable by pooling a modest lottery across cohorts/years, and is further blunted whenever the claimed effect is large (large effects are detectable at small N). Its irreducible limit is that it measures an effect only for the self-selected applicant pool. Together these make a *sparse but multi-year* lottery of a defined subgroup both feasible and interpretable.
- **Link to source:**
<https://pricetheory.uchicago.edu/levitt/Papers/schoolchoicelottery.pdf> |
<https://users.nber.org/~dynarski/w15740.pdf> |
<https://cowles.yale.edu/sites/default/files/2022-09/d2080.pdf> |
<https://journals.sagepub.com/doi/10.1177/23328584241231933>

Category 4: The Motivation Hypothesis — Can Admission Select on Drive Instead of Prior Achievement?

- **Subcategory 4.1: Grit as a Selection Signal — the case for**
 - **Source:** Duckworth et al. (2007)
 - **DOK 1 — Facts:**
 - Duckworth, A., Peterson, C., Matthews, M., & Kelly, D. (2007), "Grit: Perseverance and Passion for Long-Term Goals," *JPSP*, 92(6), 1087–1101.
 - Grit accounted for ~4% of variance in success outcomes; predicted West Point Beast Barracks retention (1 SD higher grit → OR $\approx$ 1.62) and Spelling Bee performance, over and above IQ/SAT.
 - Grit–SAT correlation was slightly *negative* ($r \approx -.20$): higher-scoring students were marginally *less* gritty.
 - **DOK 2 — Summary:** Duckworth's case is that a non-cognitive trait predicts high-stakes achievement independently of ability — and that the very highest testers are not the grittiest. If true, motivation is a legitimate, ability-orthogonal admissions signal, which is the foundation our thesis needs.
 - **Link to source:** <https://pubmed.ncbi.nlm.nih.gov/17547490/>
- **Subcategory 4.2: Grit — the skeptical counter-evidence**
 - **Source:** Credé, Tynan & Harms (2017)
 - **DOK 1 — Facts:**
 - Credé, M., Tynan, M., & Harms, P. (2017), "Much Ado About Grit: A Meta-Analytic Synthesis of the Grit Literature," *JPSP*, 113(3), 492–511 (584 effect sizes; 88 samples; ~66,807 individuals).
 - Grit–academic-performance $\rho \approx .18$ (weak); grit–conscientiousness $\rho \approx .84$ (grit is largely a relabeling of conscientiousness).
 - Only the *perseverance-of-effort* facet ($\rho \approx .26$) predicted performance; *consistency-of-interest* ($\rho \approx .10$) added little; grit interventions likely have only weak effects.
 - **DOK 2 — Summary:** The largest synthesis finds grit is a weak predictor and barely distinct from an existing personality trait, and that only its effort component carries signal. For a motivation-based admissions rule, this is the load-bearing warning: a self-reported grit score would likely be too unreliable and too collinear with conscientiousness to defend — revealed effort may be the only admissible version.

- **Link to source:** <https://pubmed.ncbi.nlm.nih.gov/27845531/>
- **Subcategory 4.3: Growth Mindset — the effect concentrates in the "behind" students**
 - **Source:** Dweck / Blackwell et al. (2007); Sisk et al. (2018); Yeager et al. (2019)
 - **DOK 1 — Facts:**
 - Blackwell, L., Trzesniewski, K., & Dweck, C. (2007), "Implicit Theories of Intelligence Predict Achievement Across an Adolescent Transition," *Child Development*, 78(1), 246–263 — growth theory predicted an upward math trajectory; an intervention reversed a control-group grade decline.
 - Sisk, V., et al. (2018), "To What Extent and Under Which Circumstances Are Growth Mind-Sets Important to Academic Achievement? Two Meta-Analyses," *Psychological Science*, 29(4), 549–571 — overall mindset-achievement $r \approx .10$; intervention $d \approx 0.08$; but low-SES students $d \approx 0.34$ and academically high-risk students $d \approx 0.19$.
 - Yeager, D., et al. (2019), "A national experiment reveals where a growth mindset improves achievement," *Nature*, 573, 364–369 (N=12,490; 65 schools) — lower-achieving students gained +0.10 GPA points (95% CI [0.04, 0.16]); higher achievers ~ 0 ; advanced-math enrollment rose ~ 3 pp.
 - **DOK 2 — Summary:** Even the skeptical meta-analysis and the large national trial converge on a pattern that cuts *in favor* of our thesis: motivation-relevant interventions do little for high achievers but meaningfully help lower-achieving and disadvantaged students. The population where the effect is real is precisely the "behind" population GT School's ability screen excludes.
 - **Link to source:** <https://doi.org/10.1111/j.1467-8624.2007.00995.x> | <https://pubmed.ncbi.nlm.nih.gov/29505339/> | <https://www.nature.com/articles/s41586-019-1466-y>
- **Subcategory 4.4: Ability's Predictive Validity & the Catch-Up Mechanism**
 - **Source:** Schmidt & Hunter (1998); Robertson et al. (2010); Bloom (1984); Kulik, Kulik & Bangert-Drowns (1990)
 - **DOK 1 — Facts:**
 - Schmidt, F., & Hunter, J. (1998), *Psychological Bulletin*, 124(2), 262–274 — general mental ability is among the strongest predictors of performance (validity $\approx .51$).

- Robertson, K., Smeets, S., Lubinski, D., & Benbow, C. (2010), "Beyond the Threshold Hypothesis," *Current Directions in Psychological Science*, 19(6), 346–351 — ability *and* motivation/commitment are both independently necessary; ability does not cease to matter above a threshold.
- Bloom, B. (1984), "The 2 Sigma Problem," *Educational Researcher*, 13(6), 4–16 — one-to-one tutoring with mastery learning ≈ 2 SD above conventional instruction; group mastery learning alone ≈ 1 SD. **Caveat (important):** Bloom's 2-sigma figure has **never been replicated** — it rests on small dissertation studies; realistic causally-identified tutoring effects are **$\sim 0.30\text{--}0.42$ SD** (Nickow, Oreopoulos & Quan 2024, meta-analysis of 96 RCTs; Cohen, Kulik & Kulik 1982 ≈ 0.33 SD). Treat "2 sigma" as an aspirational ceiling, not an expected effect.
- Kulik, Kulik & Bangert-Drowns (1990), "Effectiveness of Mastery Learning Programs: A Meta-Analysis," *Review of Educational Research*, 60(2), 265–299 — mastery learning raised exam scores ≈ 0.52 SD on average, with *stronger* effects for weaker students. (Note: mostly matched/observational, not RCTs — see the tutoring-RCT benchmark above for the causally-identified figure.)
- **DOK 2 — Summary:** Two facts sit in tension: ability is a genuinely strong predictor (so a motivation-only bet is risky), yet intensive individualized instruction — the mechanism GT School actually uses — produces its largest gains for weaker students. The realistic causal magnitude is $\sim 0.3\text{--}0.4$ SD (not Bloom's unreplicated 2 SD), and the RCT base is drawn mostly from *struggling* students, so the effect for high-ability learners is a reasonable inference rather than a tested result. The design question is whether GT School's program is the *kind* of treatment for which the behind-but-motivated student is the highest-return population — a question only a purpose-built evaluation can answer.
- **Link to source:** <https://doi.org/10.1037/0033-2909.124.2.262> | <https://doi.org/10.1177/0963721410391442> | <https://doi.org/10.3102/0013189X013006004> | <https://doi.org/10.3102/00346543060002265>

- **Subcategory 4.5: If Not "Motivation," What Does Predict Growth? Prior Ability + Conscientiousness**
 - **Source:** Guez et al. (2018); McManus et al. (2013); Poropat (2009); Mammadov (2021); Duckworth & Seligman (2005); Meyer et al. (2024)
 - **DOK 1 — Facts:**
 - **Prior ability keeps predicting, with no ceiling at the top.** Guez, Peyre, Le Cam, Gauvrit & Ramus (2018), *Intelligence* — in a representative French sample (**N=30,489**), the grade-6-IQ→grade-9-achievement slope barely changes when restricted to the top 2% ($\beta=.11$) or top 5% ($\beta=.15$); the relationship is essentially *linear*, with no threshold. McManus et al. (2013), "The Academic Backbone," *BMC Medicine* — prior attainment is the dominant autoregressive predictor (within-school year-to-year correlations $\sim 0.73\text{--}0.75$). (Cf. SMPY's within-top-1% gradients in 2.3.)
 - **The measurable, ability-*independent* lever is conscientiousness/self-discipline — not grit or mindset.** Poropat (2009), *Psychological Bulletin* ($k=138$, $N=70,926$): conscientiousness is the strongest Big Five predictor of achievement (corrected $r=.22$), rising to partial $r=.24$ controlling for intelligence, and it adds prediction (partial $r=.17$) even *after* controlling for prior GPA. Mammadov (2021) (267 samples, $N=413,074$): conscientiousness corrected $p=.27$. Duckworth & Seligman (2005), *Psychological Science*: self-discipline accounted for **>2× the variance of IQ** in adolescents' grades, incrementally after prior grades, achievement tests, and IQ.
 - **Conscientiousness appears to *synergize* with ability at the high end** (Meyer et al. 2024, *European Journal of Personality*, $N=18,637$: small interaction *amplifying* the ability→achievement link) — though this literature is mixed (compensatory effects found elsewhere).
 - **DOK 2 — Summary:** The honest answer to "if not motivation, what?" is: **prior ability/achievement dominates** (and does not plateau at the top — a genuine check on any pure motivation bet), while the best-validated *ability-independent* lever is **conscientiousness/self-discipline**, the measurable trait that "grit" was a noisy proxy for. This refines rather than refutes the thesis: the defensible pivot is to select on **demonstrated prior attainment + ability breadth (including the spatial dimension in 1.6) + conscientiousness**

measured behaviorally (cf. the PERC task) — not on fuzzy self-reported "motivation," and not on a narrow verbal/quant content cutoff alone. **Honest caveat:** the conscientiousness evidence is drawn from general/mixed-ability samples; whether it retains incremental validity *within* an already-top-percentile cohort (under range restriction) is an open question, not an established result.

- **Link to source:** <http://www.lscpl.net/persons/ramus/docs/INTELL18B.pdf> | <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3827330/> | <https://gwern.net/doc/psychology/personality/conscientiousness/2021-mammadov.pdf> | <https://journals.sagepub.com/doi/abs/10.1111/j.1467-9280.2005.01641.x>

Category 5: GT School / Alpha — The Program and the Selection Critique

- **Subcategory 5.1: What the Program Is**
 - **Source:** Alpha School / 2 Hour Learning marketing [COMPANY CLAIM]; Reason (Segan, 2026) [INDEPENDENT]; Wikipedia [INDEPENDENT]
 - **DOK 1 — Facts:**
 - Model: ~2 hours/morning of app-based adaptive "core" academics; afternoons for life-skills workshops. Adults are "guides," reportedly prohibited from direct instruction. [COMPANY CLAIM / INDEPENDENT]
 - Founded 2014 in Austin, TX (spun off from Acton Academy); public face MacKenzie Price; funded by Joe Liemandt. "Timeback" is the rebranded platform (marketed as a "\$100M" system to "learn 10x faster"). [INDEPENDENT]
 - Founders acknowledge the "AI" is adaptive-learning software (IXL/Khan-style), not an LLM chatbot; the math engine has shifted (IXL → Math Academy/TimeBack). [INDEPENDENT]
 - GT School is the gifted branch, near Austin (Georgetown), same model; Pamela Hobart is its gifted "evangelist." [INDEPENDENT]
 - **Alpha vs. GT School — same model, different audience:** Alpha serves a *broad* ability range (MacKenzie Price describes catching students up "no matter where they are... 10th percentile or 98th percentile"), whereas GT School is the *gifted-only* branch, screening applicants with the CogAT at ~90th percentile (5.3). Same 2-hour/Timeback platform, different admitted

population — so GT School's differentiation from its own sister school is precisely its exclusive focus on the top tier. [INDEPENDENT / COMPANY interview]

- **DOK 2 — Summary:** GT School runs a compressed, software-paced academic model with a for-profit private-school structure, and its underlying technology is adaptive courseware rather than novel AI. The platform has changed over time, which matters because the published results predate the current software stack. Its distinction from sister school Alpha is audience, not method: Alpha spans a broad ability range while GT School admits only the gifted tier.
- **Link to source:** <https://reason.com/2026/06/24/g-t-schools-bet-on-gifted-ed-cash-rewards-2-hours-of-ai-tutoring-no-lectures/> | https://en.wikipedia.org/wiki/Alpha_School
- **Subcategory 5.2: The Outcome Claims (self-reported)**
 - **Source:** Alpha School program page [COMPANY CLAIM]; Reason (Segan, 2026)
 - **DOK 1 — Facts:**
 - "On average, Alpha students grow 2.6 times faster than peers on nationally normed MAP tests"; "the majority of students consistently outperform national averages"; "best students achieve up to 6.5x growth." [COMPANY CLAIM]
 - Marketing invokes "top 1–2% nationally" and a comparison to the "top 0.1% of schools globally" (Exeter/Dalton/Trinity) — the latter generated by feeding a MAP dataset into LLMs, not an audit. [COMPANY CLAIM]
 - Benchmark is NWEA MAP Growth (3x/year), which Alpha describes as an independent third-party test.
 - No public "MIT-level by 8th grade" claim was found in GT School's materials; the operative elite framing is the "top 0.1% / top 1–2%" comparison. (The "MIT by 8th grade" target in this project is our own operational definition, not a GT School quote.)
 - **DOK 2 — Summary:** Every headline outcome claim is an *internal analysis* of MAP data; the test instrument is third-party but the analysis is not audited, and the flagship "elite" comparison is marketing-generated. The specific

elite-readiness bar this project targets is our own operational definition rather than a GT School claim.

- **Link to source:** <https://alpha.school/the-program/> | <https://reason.com/2026/06/24/g-t-schools-bet-on-gifted-ed-cash-rewards-2-hours-of-ai-tutoring-no-lectures/>
- **Subcategory 5.3: Admissions / Selection (the crux)**
 - **Source:** Reason (Segan, 2026) [INDEPENDENT]; NEPC/First Fish (Cherkin & Champney, 2026) [INDEPENDENT, opinion]
 - **DOK 1 — Facts:**
 - GT School screens applicants with the **CogAT** aptitude test at roughly the **90th-percentile** cutoff — an explicit cognitive-ability admissions filter. [INDEPENDENT]
 - Tuition: in-person GT School ≈ \$25,000/yr; remote "G.T. Anywhere" priced near the ~\$10,000 Texas voucher. Alpha schools range ≈ \$10,000–\$75,000/yr (most ≈ \$40,000). [INDEPENDENT]
 - As for-profit private schools they choose whom to admit; began serving mainly wealthier families, later expanded to some lower-income areas (e.g., Brownsville). [INDEPENDENT]
 - **DOK 2 — Summary:** GT School selects on measured aptitude *and* on families able to pay a substantial tuition — a double filter on exactly the variables (ability and family investment) that a skeptic would name as the source of any apparent effect. This documented CogAT cutoff is simultaneously the strongest evidence for the selection-effect worry and the natural running variable for a regression-discontinuity design.
 - **Link to source:** <https://reason.com/2026/06/24/g-t-schools-bet-on-gifted-ed-cash-rewards-2-hours-of-ai-tutoring-no-lectures/> | <https://nepc.colorado.edu/blog/fprice-kids-pay>
- **Subcategory 5.4: The Public Selection Critique & the Evidence Gap**
 - **Source:** Kane (2026) [INDEPENDENT]; Alexander/ACX [INDEPENDENT]; NEPC/First Fish (2026) [INDEPENDENT]
 - **DOK 1 — Facts:**

- Dylan Kane (Feb 2026): Alpha should compare students "to their own students' learning before they join, not to the national median"; \$40k-tuition families are "probably already learning faster than the median"; credits fast learning but attributes it to "motivation, not the technology."
- Scott Alexander (ACX) frames the debate as "affluent, selection-effect-propped program" vs. "joyless speed-drilling," noting public evidence is "thin in both directions."
- Consensus across independent commentators: the growth/percentile claims rest entirely on internal MAP analyses, have not been independently verified, and there is no peer-reviewed study, control group, or matched comparison. Pennsylvania rejected an affiliated charter as "untested."
- **DOK 2 — Summary:** Independent observers converge on the same two gaps: GT School's results lack a credible counterfactual (no baseline-relative or matched comparison, no control group, no third-party evaluation), and the most compelling alternative explanation on offer is student motivation and family selection rather than the program. Closing exactly this gap is the deliverable this BrainLift exists to design.
- **Link to source:** <https://fivetwelvethirteen.substack.com/p/alpha-schools-secret-sauce> | <https://www.astralcodexten.com/p/your-review-alpha-school> | <https://nepc.colorado.edu/blog/fprice-kids-pay>
- **Subcategory 5.5: A Future Possibility — The Texas Voucher (ESA/TEFA) Shift**
(speculative; not yet implemented for GT School, no data)
 - **Source:** Texas Tribune (2026); Texas Education Agency; Reason (Segan, 2026) — *forward-looking; all program-effect implications are speculation pending data*
 - **DOK 1 — Facts (program facts vs. speculation, kept separate):**
 - *Fact:* Texas's Education Savings Account program (TEFA) is set to launch for the **2026–27 school year** at roughly **\$10,900/student** (~85% of state per-pupil funding), up to \$30,000 for students with disabilities and \$2,000 for homeschoolers; ~\$1B first-biennium cap (~80–100k students); priority to below-poverty and disabled students, then higher incomes; **the state program itself uses a lottery when demand exceeds capacity.**
 - *Fact:* Alpha/GT appear eligible (after a Cognia-accreditation block was resolved in early 2026), and GT's remote "GT Anywhere" tier was already

priced near the ~\$10k voucher amount.

- *Speculation (no data yet)*: whether GT participates at scale, and how a voucher would change its applicant pool or selection dynamics, is **unobserved**. This subcategory records a scenario to monitor — not a present condition, and not evidence.
- **DOK 2 — Summary**: *If* GT participates at scale — and this has not happened yet — a voucher could, speculatively, weaken the **tuition** filter (largely for the ~\$10k remote tier; only partially for in-person at ~\$25k or Alpha at ~\$40k), and because the state program lotteries oversubscribed demand it could make an admission-lottery evaluation more feasible or even naturally occurring. Two limits keep this a *possibility*, not a fix: (1) a voucher removes **price**, not the **CogAT ability screen** or the **self-selection on parental engagement/information** — voucher applicants remain a self-selected, more-advantaged pool; and (2) none of it has occurred, so any claim about its effect on the selection problem is speculation. The one durable point worth flagging now: *were* it to materialize, the binding constraint on a credible evaluation would shift from tuition to the CogAT cutoff, and public funding would sharpen the case for an independent, pre-registered evaluation.
- **Link to source**: <https://www.texastribune.org/2026/05/04/texas-launches-school-vouchers-esa-choice/> | <https://tea.texas.gov/about-tea/news-and-multimedia/correspondence/taa-letters/senate-bill-2-education-savings-accounts-and-children-with-disabilities>
- **Subcategory 5.6: Why the Curated Cohort Plausibly Matters — the Pace Mechanism (and where GT School's self-paced model already neutralizes it)**
 - **Source**: Reis et al. (1998); Kulik & Kulik (1992); Steenbergen-Hu et al. (2016); Duflo, Dupas & Kremer (2011); the elite-illusion/self-paced findings
 - **DOK 1 — Facts**:
 - **Gifted students waste large amounts of time on already-mastered content**: Reis, Westberg, Kulikowich & Purcell (1998), *Gifted Child Quarterly* — teachers eliminated **40–50% of the regular curriculum** for 336 high-ability students in mixed-ability classes with **no loss** in achievement (out-of-level tested to avoid ceilings).

- **The gain comes from pace/level, not peer sorting:** between-class grouping with no curriculum change $\approx$ null ($g \approx 0.04-0.06$), but **acceleration** $\approx g=0.70$ vs. same-age peers (Steenbergen-Hu et al. 2016, in 3.4); Kulik & Kulik (1992) — effects scale with the amount of curriculum/pace adjustment (accelerated gifted $\approx \sim 1$ grade-level gain).
- **Cleanest causal support for the compromise-pace problem:** Duflo, Dupas & Kremer (2011), *AER*, randomized across **121 schools** — mixed-ability classes force a compromise pace; tracking by readiness let instruction run at the group's level and helped **both** ends ($\sim +0.14-0.18$ SD).
- **But GT School's self-paced software largely neutralizes this at the academic core:** because adaptive/mastery software paces each student individually, a fast student's rate is largely insulated from peers (the elite-illusion nulls show near-zero peer-composition effect on achievement — 3.1). The pace mechanism therefore bites mainly in the **shared components** — group workshops (Duflo) and finite guide attention — not the software.
- **DOK 2 — Summary:** There is a real, evidence-backed reason a homogeneous high-readiness cohort could accelerate gifted students: mixed-ability *group* instruction forces a compromise pace that wastes their time (Reis: 40–50% wasted; Duflo RCT), and the largest gains come from a faster, more advanced pace (acceleration $g=0.70$). Crucially, GT School's *self-paced software* was built to decouple pace from peers — so within the academic core, admitting behind-but-capable students should barely slow the fastest. The genuine cost of a more heterogeneous cohort is confined to the **shared components** (workshops, guide time), which are design variables that can be protected (readiness-grouped workshops, ring-fenced fast-track guide time) rather than reasons to keep the whole screen strict.
- **Link to source:**
<https://journals.sagepub.com/doi/10.1177/001698629804200206> |
<https://journals.sagepub.com/doi/10.1177/001698629203600204> |
<https://www.aeaweb.org/articles?id=10.1257/aer.101.5.1739>

Category 6: The Equity Blind Spot — How Aptitude Screening Encodes Cultural, Racial, and Class Disparity

Where Category 1 asks "does the screen miss able kids?", this category asks "which kids, systematically, and why?" It connects to 1.5 and 3.3 (universal screening as a partial remedy). The tension is deliberate: the same objective test that is more equitable than biased referral (1.5) is also correlated with the very advantages it claims to look past.

- **Subcategory 6.1: Racial/ethnic underrepresentation in ability-identified gifted programs**
 - **Source:** NAGC (federal OCR/CRDC data); Grissom & Redding (2016)
 - **DOK 1 — Facts:**
 - NAGC, citing 2017–18 federal Office for Civil Rights data: Black and Hispanic students are represented in gifted programs at only ~55% and ~77% of their share of the K–12 population, while Asian American and White students are at ~196% and ~108%; historically underrepresented groups are underserved "by at least 50%."
 - Education Next analysis of 2017–18 CRDC: ~6.9% of students are in gifted programs; 27.3% of gifted students are Black or Hispanic vs. 47.7% of overall enrollment.
 - Grissom, J. A., & Redding, C. (2016), "Discretion and Disproportionality," *AERA Open*, 2(1) — using ECLS-K (>10,000 students): at identical math/reading achievement, Black students were assigned to gifted programs about *half* as often as White peers; overall, Black students were 66% and Hispanic 47% less likely to be identified; Black students were ~3× more likely to be identified when taught by a Black teacher.
 - **DOK 2 — Summary:** Gifted programs identified through ability/achievement screening substantially under-enroll Black, Hispanic, and Native students relative to their population share, and the gap persists even among students with *identical* measured achievement. That residual indicates the disparity is not merely a difference in measured ability but in who gets identified and served.
 - **Link to source:** <https://www.nagc.org/identification> | <https://journals.sagepub.com/doi/full/10.1177/2332858415622175>

- **Subcategory 6.2: Cultural loading and the "culture-fair test" myth**
 - **Source:** Lohman (2005, GCQ); Lohman, Korb & Lakin (2008); historical origins of aptitude testing
 - **DOK 1 — Facts:**
 - Lohman, D. F. (2005), "The Role of Nonverbal Ability Tests in Identifying Academically Gifted Students," *Gifted Child Quarterly*, 49(2), 111–138 — nonverbal ("culture-fair") tests are "neither culture free nor culture fair"; selecting on them alone would exclude the majority of the most academically accomplished students in every ethnic group.
 - Lohman, D. F., Korb, K. A., & Lakin, J. M. (2008), *Gifted Child Quarterly*, 52(4), 275–296 — across the Raven, NNAT, and CogAT (1,198 children, ~40% ELL), ELL children scored ~0.5–0.67 SD (8–10 standard-score points) lower on all three *nonverbal* tests, and none predicted ELL achievement well — the "culture-fair" remedy did not close the gap.
 - Historical framing (contested): U.S. mass aptitude testing traces to the WWI Army tests and to figures (Terman 1916; Brigham 1923, who then developed the SAT) who explicitly tied testing to racial hierarchy. Brigham later repudiated that premise and the College Board has disavowed the legacy; critics nonetheless argue the instruments still encode dominant-culture norms.
 - **DOK 2 — Summary:** Even the tests marketed as bias-reducing leave large gaps for English-language learners and culturally diverse students, and the field's own leading psychometrician holds that no single test is culture-fair. The historical record shows aptitude testing was, at inception, entangled with claims of racial hierarchy — a contested but relevant backdrop to using such tests as gatekeepers.
 - **Link to source:**

<https://journals.sagepub.com/doi/abs/10.1177/001698620504900203> |

<https://journals.sagepub.com/doi/10.1177/0016986208321808>
- **Subcategory 6.3: Class/socioeconomic disparity — and the compounding of test + tuition**
 - **Source:** SES–cognition literature (von Stumm et al.); Card & Giuliano (2016); reasoned synthesis

- **DOK 1 — Facts:**

- SES is among the most robust predictors of children's cognitive-test performance: highest- vs. lowest-SES children differ by ~6 IQ points at age 2, a gap that nearly triples by age 16 (von Stumm et al.); vocabulary shows the strongest SES association of any cognitive domain.
- Card & Giuliano (2016): under the traditional referral system, only 28% of gifted 3rd-graders were Black or Hispanic vs. 60% of district enrollment; low-income and ELL students were systematically under-referred.
- **Correction / honesty note:** the "test prep compounds the advantage" mechanism is **weaker than commonly assumed**. Causally-identified estimates put commercial coaching at only ~**20 SAT points** after controlling for selection (Briggs, NELS:88) and ~**+9 percentile** (federal What Works Clearinghouse) — far below marketing claims — and **no CogAT-specific coaching evidence exists**. So the SES→score link runs mainly through the **developmental** gap (the ~6-IQ-point gap widening through adolescence), not through test prep.
- (*Reasoned synthesis, not a single cited statistic*) Even setting coaching aside, when a program charges tuition (GT School ≈ \$25k) or gates scholarships on the same aptitude test, the aptitude filter and the ability-to-pay filter — both correlated with SES — compound.

- **DOK 2 — Summary:** Cognitive-test scores are strongly correlated with socioeconomic status from before school entry, so an aptitude cutoff functions partly as a proxy for family advantage — but that proxy operates chiefly through the *developmental* SES–score gap, not through coachable test prep (whose measured effect is small and unstudied for CogAT). A program that also charges substantial tuition applies a second, SES-correlated filter, concentrating access among the already-advantaged unless the design deliberately counteracts it.
- **Link to source:** <https://pmc.ncbi.nlm.nih.gov/articles/PMC4641149/> | <https://www.pnas.org/doi/10.1073/pnas.1605043113>

- **Subcategory 6.4: Frameworks and remedies**

- **Source:** Perez (2023, JHU "Tale of Two Cities"); NAGC equity position; Card & Giuliano (2016)

- **DOK 1 — Facts:**
 - Perez, R. (2023), "A Tale of Two Cities...and Their Gifted Education Systems," Johns Hopkins *Hopkins Insider* (an undergraduate research write-up, not peer-reviewed) — comparing Baltimore and Singapore, argues both systems "responded to gifted minorities by denying their existence," and that culturally specific definitions of giftedness advantage dominant groups (invoking the involuntary- vs. voluntary-minority distinction).
 - NAGC equity position ("Identifying and Serving Culturally and Linguistically Diverse Students"): "giftedness ... is found among all racial, ethnic, and income groups"; "one test at a specific point in time should not dictate whether someone is identified"; recommends universal screening, multiple measures, local norms, and front-loading/talent development.
 - Card & Giuliano (2016) is the empirical proof-of-concept that a design change (universal screening) can materially reduce the disparity without lowering the eligibility bar.
- **DOK 2 — Summary:** The equity literature converges on a prescription that mirrors this project's design question: single-point, single-test gatekeeping is the problem, and universal screening plus multiple measures and local norms is the evidence-based remedy. The disparity critique is therefore not an argument against selection but an argument for redesigning it.
- **Link to source:** <https://apply.jhu.edu/hopkins-insider/a-tale-of-two-cities-and-their-gifted-education-systems/> | <https://www.nagc.org/equity> | <https://www.pnas.org/doi/10.1073/pnas.1605043113>

Appendix: Citation Verification Notes

Load-bearing citations were verified against primary sources (journal pages, PubMed, NBER, PNAS). A few items warrant a final check before being quoted to an external expert:

1. **SMPY top-0.01% outcome percentages (44% doctorates / ~15% patent / >11% tenure, N=320):** reported by two independent research passes citing Kell, Lubinski & Benbow (2013); the exact figures should be confirmed against the 2013 article body before being quoted to the decimal.
2. **"Elite illusion" (Abdulkadiroğlu, Angrist & Pathak 2014):** the null-at-the-margin finding is verified; a *verbatim* one-sentence definition of the phrase was not located — the phrase should be paraphrased rather than quoted.
3. **Bui, Craig & Imberman:** the published *AEJ: Economic Policy* subtitle ends "...on Students"; the earlier NBER WP 17089 ends "...on Achievement." Exact per-subject SD coefficients were not verified to the digit; the qualitative nulls are.
4. **Card & Giuliano — two distinct papers:** the 2016 *PNAS screening* paper (Cat. 3.3) is fully verified. The separate gifted-*classroom* achievement-effects paper — "Can Tracking Raise the Test Scores of High-Ability Minority Students?", *AER* 106(10) (Cat. 3.7) — was verified from the primary PDF (Table 3 effect sizes). Keep the two straight; they are different studies. 4b. **Booij, Haan & Plug (Cat. 3.7):** the 2016 and 2017 items are **IZA working papers**; confirm final peer-reviewed publication (and the 2017 per-subject SEs) before load-bearing citation. The 2016 GPA effects (+0.38/+0.30/+0.44 SD) were verified from the DP PDF.
5. **Kelley's regression-to-the-mean formula:** the attribution is standard; the verified peer-reviewed anchor is Maassen (2000), *Psychometrika*.
6. **Category 1 (CogAT-too-strict) figures:** the Lohman (2005) 32%/18% identification rates and the Lohman & Korb (2006) "70% no longer top-3% by grade 8" figure were extracted verbatim from the primary PDFs; the Ramsden (2011) IQ-change figures are fully verified against *Nature*; the McBee et al. (2016) ">60% false-negative" is verified from the article. *Achievement Trap* figures: the "3.4 million" was confirmed directly from the report text, and the fifth-grade fall-out rates (56%/69% retained; ~7% vs. 16–17% rising in) were confirmed via Fordham Institute and Education Week summaries — worth a final check against the primary report's page numbers before formal citation.
7. **GT School facts:** [COMPANY CLAIM] items (self-reported MAP analyses, "2.6x," "top 0.1%") are distinguished from [INDEPENDENT] reporting (CogAT cutoff, tuition, the critiques). No public "MIT-level by 8th grade" claim was found; that target is our own operational definition.

8. **Co-founder / byline notes:** the Reason (2026) article byline is Shreedha Segar; the Alpha co-founder is variously reported as Andrew Price (marketing/journalism) or Brian Holtz (Wikipedia).
9. **The Joe Liemandt / CogAT attribution (subcategory 1.5):** *no documented statement from Joe Liemandt or MacKenzie Price crediting CogAT specifically with a positive impact was found.* The "~90th-percentile CogAT cutoff" originates in the interviewer's framing in the Reason (2026) piece; the on-record pro-screening voice is GT staffer Pamela Hobart. If a primary source for Joe's claim exists (interview/podcast), insert it verbatim; until then it should not be attributed to him.
10. **Counterargument figures (1.5):** Deary et al. (2007) is $r \approx .69$ observed / $.81$ latent — cite the observed figure as the conservative one. Schmidt & Hunter's GMA validity ($.51$) is contested by Sackett et al. (2022), who argue the corrected estimate is nearer $\sim .31$; note this if cited. Grissom & Redding exact percentages come from Vanderbilt's official research summary (confirm against the SAGE full text).
11. **Category 6 figures:** NAGC "55%/77% vs. 196%/108% of expected representation" (2017–18 OCR) is the primary-anchored national stat; the "56%/68%" OCR split circulating elsewhere is secondary. The SES "6 IQ points at age 2 → nearly triples by 16" figure is from the von Stumm work (PMC4641149) — confirm before formal citation. **Test-prep correction (6.3):** commercial coaching effects are *small* (~ 20 SAT points after selection controls; $\sim +9$ percentile, WWC) and there is **no CogAT coaching evidence** — so the SES→score link is attributed mainly to the developmental gap, not test prep; the tuition/scholarship compounding is reasoned synthesis. The JHU "Tale of Two Cities" is an undergraduate admissions-blog piece, not peer-reviewed — cite as framing only.
12. **Catch-up mechanism (4.4):** Bloom's "2-sigma" is **not replicated**; the causally-identified tutoring benchmark is ~ 0.29 – 0.42 SD (Nickow, Oreopoulos & Quan 2024, 96 RCTs; Cohen/Kulik/Kulik 1982). Tutoring RCTs are drawn mostly from *struggling* students, so the high-ability transfer is an inference, not a tested result.